

Fatih Berkay Sarpkaya

Education

- 2022–Present **Doctor of Philosophy (Ph.D.)**, *New York University Tandon School of Engineering*, Brooklyn, NY
Expected Graduation: May 2027
Major: Electrical & Computer Engineering | Advisor: Professor Shivendra Panwar | **GPA: 3.949/4.00**
- 2017–2022 **Bachelor of Science (B.Sc.)**, *Bilkent University*, Ankara, Turkey
Major: Electrical and Electronics Engineering, **CGPA : 3.77/4.00**

Research Interests

Transport Protocols & Congestion Control | Low-Latency Networking | Data Center Networking

Work & Research Experience

- Summer 2026 **Microsoft Research**, *AI Network Observability Research Intern*, Redmond, WA
Performance Modeling and Analysis of AI Accelerator Infrastructure — Managers: Anqi Guo, Nicholas Lurski, Pratik Marolia
- Developed and calibrated an FPGA-based emulation framework for AI accelerator collective communication workloads.
 - Modeled communication pipelines and dependencies for collective operations, validating emulation behavior against hardware measurements.
 - Characterized collective communication performance across message sizes, identifying synchronization and communication overheads affecting end-to-end latency.
- Summer 2025 **Nokia Bell Labs**, *Network Reliability Research Intern (Summer) / Co-op (Fall)*, Murray Hill, NJ
- Fall 2025 **Performance Evaluation of Congestion Control in Large BDP Networks** — Managers: Andrea Francini, Bilgehan Erman, Gary Atkinson
- Designed and executed large-scale TCP congestion control experiments on the FABRIC testbed.
 - Optimized Linux kernel congestion control modules and integrated DPDK-based traffic management tools for high-speed traffic analysis.
- Summer 2024 **InterDigital**, *Research & Innovation Wireless Intern*, New York, NY
Flexible Packet Coding Performance Analysis & Generative Semantic Communication Literature Review — Manager: Pascal Adjakple
- Implemented various network coding and packet duplication simulations in MATLAB, and analyzed performance results.
 - Conducted a literature review on generative AI in semantic communication and its potential impact on modern communication systems.
- Sep. 2022 – Present **NYU Wireless Research Center**, *Graduate Researcher*, Brooklyn, NY
- Investigating transport protocol coexistence and low-latency networking, with a focus on L4S, BBR, and DualQ AQM.
 - Designing mechanisms for end-to-end low-latency service using SRv6/BGP signaling and traffic isolation.
 - Evaluating congestion control algorithms for large data transfers over high-bandwidth-delay-product networks.
 - Conducted additional research on 5G/6G XR systems and reliable communication under adversarial channels, including capacity analysis and LLR-based decoding.

Publications

- July 2026 **Extending Low Latency Service Across the Internet**
Harkirat Singh*, **Fatih Berkay Sarpkaya***, Hakan Gulec, Fraida Fund, Shivendra Panwar
IEEE International Conference on Network Protocols (ICNP 2026) *Equal contribution [\[Link\]](#)

- June 2026 **Inherited Fragility in Dual-Queue AQMs under Heterogeneous Congestion Control**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
Under Review
- June 2026 **Evaluation of TCP Congestion Control for Public High-Performance Wide-Area Networks**
Fatih Berkay Sarpkaya, Andrea Francini, Bilgehan Erman, Shivendra Panwar
IEEE High Performance Switching and Routing (HPSR 2026) [\[Link\]](#)
- May 2025 **BBR's Sharing Behavior with CUBIC and Reno**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
IFIP Networking 2025 [\[Link\]](#)
- March 2025 **To Adopt or Not to Adopt L4S-Compatible Congestion Control? Understanding Performance in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
Passive and Active Measurement Conference (PAM 2025) [\[Link\]](#)
- Oct. 2024 **Capacity of a Binary Channel with a Time-Bounded Adversary**
M. Ying, **Fatih Berkay Sarpkaya**, S. Bakirtas, Elza Erkip, T. Rappaport, Sundeep Rangan
58th Asilomar Conference on Signals, Systems, and Computers [\[Link\]](#)
- July 2024 **To Switch or Not to Switch to TCP Prague? Incentives for Adoption in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
Applied Networking Research Workshop (ANRW 2024) [\[Link\]](#)

Relevant Coursework and Tools

- NYU - ECE Probability & Stochastic Processes | Internet Architecture & Protocols | Network Design and Algorithms | Data Center & Cloud Computing | Machine Learning | Network Modeling and Analysis | Digital Communication | Wireless Communication | Introduction to Wireless Networking | AI/ML for Network Traffic Analysis
- Bilkent - EE Computer Networks | Telecommunications I | Telecommunications II | Probability and Statistics | Statistical Learning and Data Analytics
- Tools Python | MATLAB | Linux | FABRIC Testbed | DPDK | CloudLab | GOAL

Work Experience (Undergraduate)

- Aug. 2021 – **ASELSAN**, *Candidate Engineer / Digital Design Intern*, Ankara, Turkey
- Jun. 2022 [○](#) Implemented an edge-detection pipeline in C++ and MATLAB; developed UART and I²C communication modules on FPGA.

Awards & Honors

- 2026 **Dante Youla Award for Graduate Research Excellence**, NYU Electrical and Computer Engineering
- 2022-Present School of Engineering **Fellowship** at NYU Tandon School of Engineering
- 2017-2022 Comprehensive **Scholarship** at Bilkent University
- 2017 Ranked **450th** in **University Entrance Exam (LYS)** among 2 million high school students.

Teaching & Mentoring

- 2023–2024 **Course Assistant**, *Internet Architecture & Protocols; Communication Networks*, NYU Tandon School of Engineering
- Summer 2023 **Research Mentor**, *ARISE High School Summer Research Program*, NYU Tandon School of Engineering

Additional Activities

- 2018-2022 Active Member of Bilkent University IEEE Student Branch
- July 2019 Engaged in short-term volunteer work supporting children with disabilities through the European Voluntary Service (EVS) in Macedonia.